

Photo-electric effect

When light of sufficiently high frequency falls upon the metal surface, electrons are emitted from it. This phenomenon is known as *photoelectric effect*. The emitted electrons are called photo-electrons and the current constitutes by their electrons called *photo-current*.

OR: Ejection of electrons from a metal plate when illuminated by light or any other radiation of suitable wavelength (or frequency) is called *photo-electric effect*.

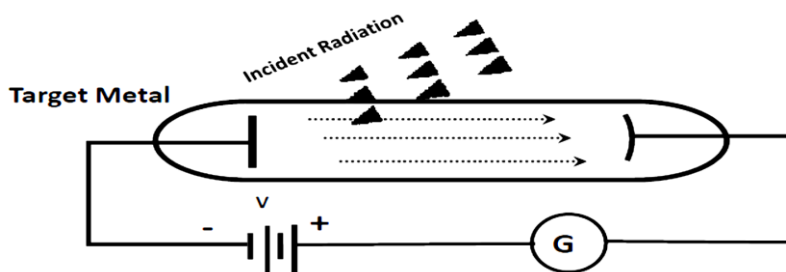


Fig. 1. Photo-electric effect

Laws of photo-electric emission

The laws of photoelectric emission are:

- (i) There is no time gap between the incidence of light and emission of electrons. *i.e., photo-electric effect is an instantaneous phenomenon.*
- (ii) For a given metallic surface, there is a minimum frequency for which the incident light can eject the photo-electrons out of the metal. Light of frequency smaller than the particular value cannot eject electrons. The *threshold frequency* is the minimum frequency which can eject an electron out of the metal.
- (iii) The number of photo-electrons ejected depends upon the intensity of the incident light. Thus, the photo-electric current depends upon the intensity of the incident light.
- (iv) Light of frequency higher than the *critical frequency* ejects electrons of different velocities. The *maximum velocity*, with which an electron is ejected, depends upon the frequency of the incident light. The *maximum kinetic energy* of the ejected electrons also increases linearly with the frequency of the incident light.

Einstein's photoelectric equation

Einstein explained the photoelectric effect based on Planck's theory of light. According to quantum theory, radiation is regarded as quanta of photons each of energy

$$E = h\nu$$

moving in space with the velocity of light. Where, h = Planck's constant & ν = frequency of radiation.

When a photon of energy $h\nu$ is incident on a certain metallic surface, it is completely absorbed and imparts its energy to a single electron. Thus the photon energy is utilized in two purposes:

- (i) Partly for getting the electron free from the atom and away from the metal surface.
- (ii) The balance of the photon energy is used up in giving the electron a kinetic energy.

The above phenomena can be written as

$$h\nu = (KE)_{\max} + w_0 \quad (1)$$

Where, $(KE)_{\max}$ is the maximum photoelectron energy and w_0 is the work function which is the minimum energy needed for an electron to leave the metal.

Again $w_0 = h\nu_0$, Where, ν_0 is the *threshold frequency* which is the minimum frequency of the photon which can eject an electron out of the metal.

Equation (1) becomes

$$h\nu = (KE)_{max} + h\nu_0$$

$$(KE)_{max} = h\nu - h\nu_0$$

$$(KE)_{max} = h(\nu - \nu_0)$$

$$\frac{1}{2}mv_{max}^2 = h(\nu - \nu_0)$$

This is Einstein's photoelectric equation.

Stopping potential

The potential at which the photo electric current becomes zero is called *stopping potential* or *cut-off potential*.

Equation of stopping potential

Einstein's photoelectric equation is

$$(KE)_{max} = h(\nu - \nu_0) \quad (1)$$

We can write

$$(KE)_{max} = eV_0 \quad (2)$$

From equation (1) and (2) we can write

$$eV_0 = h(\nu - \nu_0)$$

$$V_0 = \frac{h}{e}(\nu - \nu_0)$$

This is the equation of stopping potential in photoelectric phenomenon.

Failure of wave theory of light to explain photo-electric emission

The failures of wave theory of light to explain photo-electric emission are discussed below:

- (i) According to wave theory, the photo-electric effect is possible at the light of high intensity of any frequency. But from the photo-electric effect it is seen that every metal has its own initial (*threshold*) frequency. Below this initial (*threshold*) frequency no electrons are emitted. *Therefore this observation cannot be explained by wave theory.*
- (ii) According to wave theory, velocity and kinetic energy the emitted electrons depends on the intensity of light. But, in photo-electric effect velocity and kinetic energy of the emitted electrons is independent of the intensity of light. *Therefore this observation cannot be explained by wave theory.*
- (iii) In photo-electric effect, there is no time interval between the arrival of light at a metal surface and emission of photo-electron. The minimum energy of photo-electrons is 1 eV . But, according to wave theory 10^8 sec , i.e., about one year is needed to require 1 eV energy by the photo-electrons. *Therefore this observation cannot be explained by wave theory.*

From the above discussion it is clear that, *the wave theory of light is failed to explain photo-electric emission.*

Mathematical Problems

Problem-1: A photoelectric surface has a work function of 4 eV. What is the maximum velocity of Photoelectrons emitted by light of frequency 10^{15} Hz incident on the surface?

Solution: We know,

$$h\nu = (KE)_{\max} + w_0$$

$$\frac{1}{2}mv_{\max}^2 = h\nu - w_0$$

$$v_{\max} = \sqrt{\frac{2(h\nu - w_0)}{m}}$$

$$v_{\max} = \sqrt{\frac{2(6.63 \times 10^{-34} \times 1 \times 10^{15} - 6.4 \times 10^{-19})}{9.1 \times 10^{-31}}}$$

$$\therefore v_{\max} = 2.24 \times 10^5 \text{ ms}^{-1} \text{ (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$w_0 = 4 \text{ eV}$$

$$= (4 \times 1.6 \times 10^{-19}) \text{ J}$$

Problem-2: Find the maximum velocity of photo-electrons emitted by radiation of frequency 3×10^{15} Hz from a photoelectric surface having a work function 4.0 eV.

Solution: Do yourself [Same as **Problem-1**]. (Ans: $1.72 \times 10^6 \text{ ms}^{-1}$)

Problem-3: The wavelength of light falling on the surface of a metal of work function 2.3 eV is 4300 Å with what velocity the electron will be emitted?

Solution: We know,

$$h\nu = (KE)_{\max} + w_0$$

$$\frac{1}{2}mv_{\max}^2 = h\left(\frac{c}{\lambda}\right) - w_0$$

$$v_{\max} = \sqrt{\frac{2\left\{h\left(\frac{c}{\lambda}\right) - w_0\right\}}{m}}$$

$$v_{\max} = \sqrt{\frac{2\left\{6.63 \times 10^{-34} \times \left(\frac{3 \times 10^8}{4.3 \times 10^{-7}}\right) - 3.68 \times 10^{-19}\right\}}{9.1 \times 10^{-31}}}$$

$$\therefore v_{\max} = 4.55 \times 10^5 \text{ ms}^{-1} \text{ (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$w_0 = 2.3 \text{ eV}$$

$$= (2.3 \times 1.6 \times 10^{-19}) \text{ J}$$

$$= 3.68 \times 10^{-19} \text{ J}$$

$$\lambda = 4300 \text{ Å}$$

$$= 4.3 \times 10^{-7} \text{ m}$$

$$v_{\max} = ?$$

Problem-4: Calculate the threshold frequency and corresponding wavelength of radiation incident on a certain metal whose work function is $3.31 \times 10^{-19} \text{ J}$. Given Plank's constant = $6.63 \times 10^{-34} \text{ JS}$.

Solution: We know,

$$w_0 = h\nu_0$$

$$\nu_0 = \frac{3.31 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\therefore \nu_0 = 5 \times 10^{14} \text{ Hz (Ans)}$$

Again, we know,

$$\lambda_0 = \frac{c}{\nu_0}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ JS}$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

$$w_0 = 3.31 \times 10^{-19} \text{ J}$$

$$\nu_0 = ?$$

$$\lambda_0 = ?$$

$$\lambda_0 = \frac{3 \times 10^8}{5 \times 10^{14}}$$

$$\lambda_0 = 6 \times 10^{-7} m$$

$$\therefore \lambda_0 = 6000 \text{ \AA} \text{ (Ans)}$$

Problem-5: The stopping potential for electrons emitted from a metal due to photoelectric effect is found to be 1V for a light of 2500 \text{ \AA}. Calculate the work function of the metal in eV.

Solution: We know,

$$v = \frac{c}{\lambda}$$

$$v = \frac{3 \times 10^8}{2.5 \times 10^{-7}}$$

$$v = 1.2 \times 10^{15} \text{ Hz}$$

Again, we know,

$$hv = (KE)_{\max} + w_0$$

$$w_0 = hv - (KE)_{\max}$$

$$w_0 = hv - eV$$

$$w_0 = 6.63 \times 10^{-34} \times 1.2 \times 10^{15} - 1.6 \times 10^{-19} \times 1$$

$$w_0 = 6.356 \times 10^{-19} J$$

$$w_0 = \frac{6.356 \times 10^{-19}}{1.6 \times 10^{-19}} eV$$

$$\therefore w_0 = 3.97 \text{ eV (Ans)}$$

Here,

$$V = 1 \text{ volt}$$

$$\lambda = 2500 \text{ \AA}$$

$$= 2.5 \times 10^{-7} m$$

$$h = 6.63 \times 10^{-34} JS$$

$$c = 3 \times 10^8 ms^{-1}$$

$$e = 1.6 \times 10^{-19} C$$

$$w_0 = ?$$

Problem-6: UV light of wavelength 350 nm and intensity 1.00 w/m² is directed at a potassium surface. Find the maximum K.E. of photoelectrons. (Work function of Potassium is 2.2 eV)

Solution: We know,

$$hv = (KE)_{\max} + w_0$$

$$(KE)_{\max} = h \left(\frac{c}{\lambda} \right) - w_0$$

$$(KE)_{\max} = 4.14 \times 10^{-15} \left(\frac{3 \times 10^8}{350 \times 10^{-9}} \right) - 2.2$$

$$\therefore (KE)_{\max} = 1.35 \text{ eV (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} JS$$

$$= \frac{6.63 \times 10^{-34}}{1.6 \times 10^{-19}} eV - S$$

$$= 4.14 \times 10^{-15} eV - S$$

$$w_0 = 2.2 \text{ eV}$$

$$\lambda = 350 \text{ nm}$$

$$= 350 \times 10^{-9} m$$

$$c = 3 \times 10^8 ms^{-1}$$

$$(KE)_{\max} = ?$$